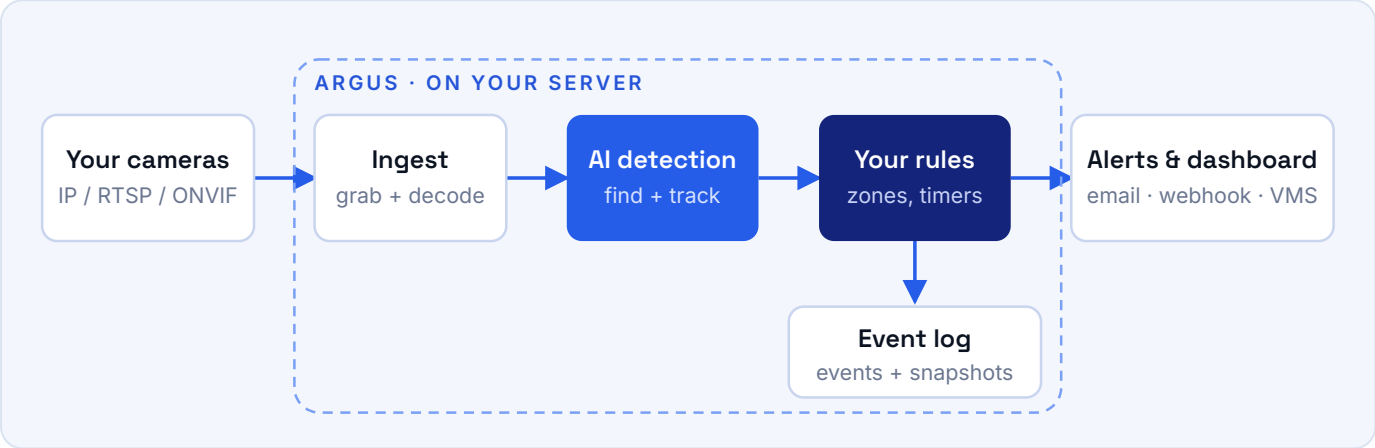
How the platform works

One AI system, installed on your site, that plugs into the cameras you have and sends clear events to the tools you already use.

ARCHITECTURE

Small parts, one pipeline



Each stage runs as its own small service linked by a fast message bus. A stage can restart or scale on its own without stopping the rest. Adding a new detection means adding a model and a rule set, not a new system.

KEY FACTS

Runs on site

A standard Linux or Windows server or VM on your network. Cloud hosting is possible on request.

CPU first

No graphics card needed to start. Add an NVIDIA GPU later for many cameras or heavier models.

Your cameras

Hikvision, Dahua, Reolink, Uniview, Axis, or any camera with an RTSP stream. Auto-discovery over ONVIF.

Proven AI models

YOLOX detectors run with ONNX Runtime. Apache-2.0 licensed, safe for commercial use.

One config per site

Cameras, zones, lines and thresholds live in a simple config file, drawn in a visual editor.

Open outputs

Every event is plain JSON, easy to send to your dashboard, database or security system.

OUTPUTS

Clear events, not raw video

The system doesn't send you footage to watch. It sends short, structured events: what happened, where, when, and a snapshot.

```
{ "type": "cross",
  "rule": "gate_line",
  "direction": "in",
  "label": "person",
  "track_id": 3,
  "camera": "gate",
  "time": "2026-10-02T02:14:07" }
```

Event types: enter, exit, dwell (stayed too long), cross (with direction).

RELIABILITY

Built to keep running

- **Auto-reconnect.** If a camera reboots or the switch loses power, the stream comes back on its own.
- **Always live.** When the server is busy, old frames are skipped so alerts never lag behind.
- **Stable tracking.** Each object keeps its ID through short blockages, so one person is one event.
- **No alert floods.** Filters and a cooldown per rule.
- **Tested code.** Automated tests cover detection, cameras and zone rules.

SIZING

Pick the model to fit your hardware

MODEL	BEST FOR	HARDWARE
Nano	Small sites, a few cameras, low-power boxes	Basic CPU, Raspberry Pi-class device
Tiny	Better accuracy, still light	Standard CPU
Small	Busy scenes, smaller or distant objects	Strong CPU or GPU
Medium / Large	Accuracy first, many cameras	NVIDIA GPU

Cameras' low-resolution sub-stream is used for detection by default, which is enough for most scenes and far lighter on the server.

SECURITY & PRIVACY

Your data stays yours

- **On your premises.** Video is processed on your server and does not need to leave your network.
- **Only events are kept.** Frames without an event are never stored.
- **Credentials protected.** Camera passwords are kept out of config files and masked in all logs.
- **You set retention.** Decide how long events and snapshots are kept.

FOR A PILOT

What we need from you

- Access to 2–5 cameras (address and login)
- A server or PC on the same network. We advise on the spec.
- One contact who knows the site
- Agreement on what should trigger an alert

ON THE ROADMAP

Web dashboard with live view · Mobile app and push alerts · Email and webhook alerts · Multi-site management · GPU acceleration · Custom models trained on your footage

Questions from your IT team?

We are happy to walk them through a live demo.

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